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COM 216 STATISTICAL COMPUTING II

SPSS usually stands for Statistical Package for the Social Sciences. It is an IBM (International Business Machines) tool that was originally launched in 1968 by SPSS Inc, but was acquired by IBM in 2009. It is one software package that various researchers mainly used for statistical data analysis and for complex statistical data analysis.

SPSS is mainly used in the following areas such as healthcare, research companies, education researchers, data miners, marketing and healthcare researchers and many others.

Features of SPSS

Here are the features of SPSS for students:

- All data in SPSS is stored in SAV format. This makes the process of processing, analyzing and extracting data very easy.
- SPSS helps you unambiguously get data from your mission-critical data. This is done through trend analysis, assumptions and predictive models, as well as some characteristics of SPSS.
- SPSS offers users deep statistical capabilities to analyze accurate results.
- It helps to get the data management system and editing tools easily
- SPSS is a tool that helps you more visually design, build, report, and present functions.
- SPSS is easy to learn, use and apply.

In addition to statistical data analysis, SPSS software also provides data management functions. It allows the user to make choices, create data, execute a file, etc. Another feature of SPSS is data documentation. Essentially, this function stores a metadata dictionary along with the data file.

This metadata vocabulary acts as a centralized repository for data-related information such as meaning, relationships with other data, origins, usage, and format.

SPSS software is used to perform quantitative analysis and complete a point-and-click statistical package.

Various statistical methods can be used in SPSS, namely:

- Descriptive statistics, including SPSS methodologies, are statistics of frequencies, cross-tabs, and descriptive relationships that are very useful.
- Two-dimensional statistics, including techniques such as analysis of variance (ANOVA), means, correlation, nonparametric tests, etc.
- Predicting various data to identify groups, including methodologies such as cluster analysis, factor analysis, etc.
- Predict a numeric result, such as linear regression.

As its name suggests, SPSS statistics software is only used to perform statistical operations.

Main functions of SPSS

SPSS offers four programs to help researchers solve complex data analysis problems.

1. Statistical program

SPSS statistics program provide many basic statistics functions, some of them include frequencies, cross-tabulation, two-dimensional statistics, Count, Sum, Do if, Do Repeat, data stacking, data restructure, weighting etc.

2. Modeler software

With SPSS Modeler software, researchers can create and validate predictive models using advanced statistical procedures.

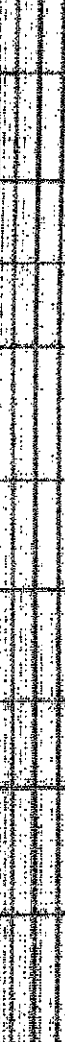
3. Text analytics software for polls This provides reliable feedback analysis, which in turn provides insight into the actual plan. SPSS Text Analytics for Surveys helps survey administrators extract valuable insights from responses to open-ended survey questions.

4. Visualization designer

The researchers found that this visual designer data makes it easy to create a wide variety of visual effects, such as density charts and radial box charts.

How does SPSS work for students?

How SPSS works.



- # Introduction to Derivatives and Risk on SPSS

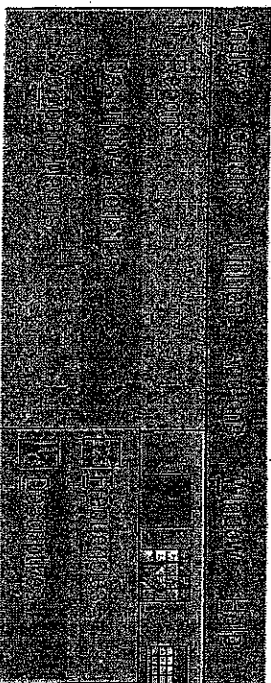
• Invalid responses

Descriptives

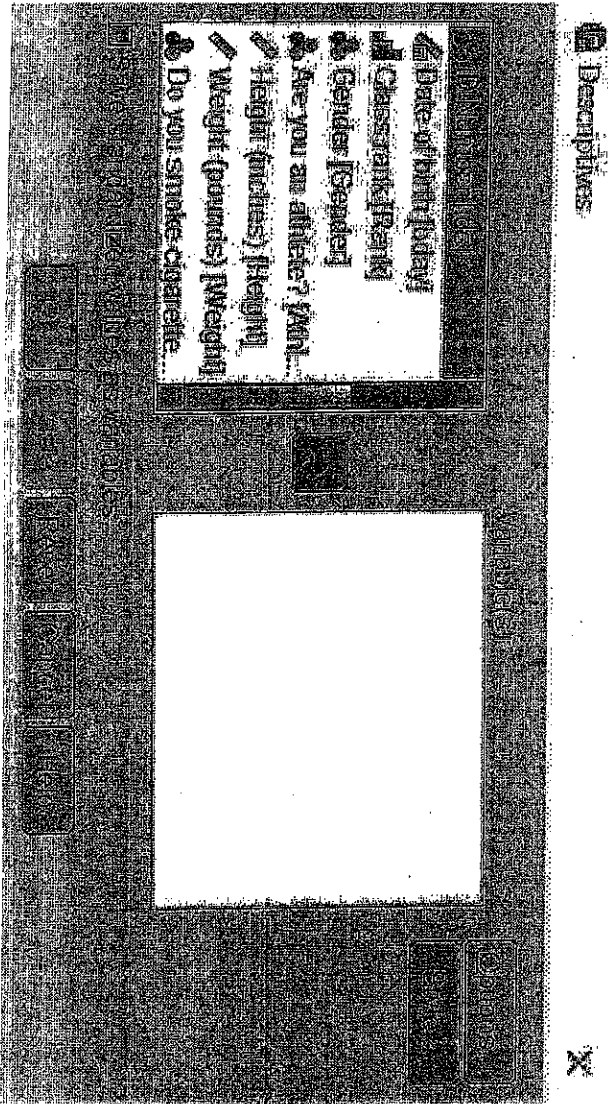
- Mean
- Sum
- Standard deviation
- Variance
- Minimum
- Maximum
- Range
- Standard error of the mean (or *S.E. mean*)
- Skewness
- Kurtosis

Additionally, the Descriptive procedure can optionally compute Z scores as new variables in your dataset.

Notice that the Descriptive procedure can't compute medians or quartiles. If you need those statistics, you'll need to use the **Frequencies**, **Explore**, or **Compare Means** procedures. The Descriptive procedure is best used when you want to compare the descriptive statistics of several numeric variable side-by-side. To run the Descriptives procedure, select **Analyze > Descriptive Statistics > Descriptive**



The Descriptives window lists all of the variables in your dataset in the left column. To select variables for analysis, click on the variable name to highlight it, then click on the arrow button to move the variable to the column on the right. Alternatively, you can double-click on the name of a variable to move it to the column on the right.

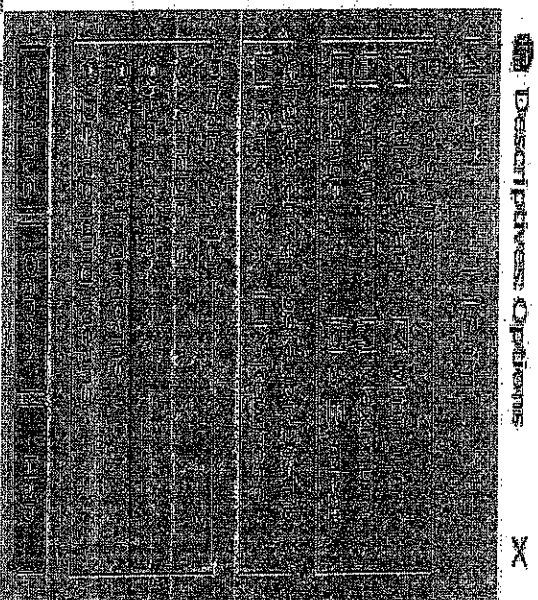


Selecting the **Save standardized values as variables** check box will compute new variables containing the standardized values (also known as *Z scores*) of each of the input variables. Recall that the standardized value of a variable is computed by subtracting its mean and then dividing that difference by the standard deviation:

$$Z = \frac{x - \mu}{\sigma}$$

By default, the Descriptive procedure computes the mean, standard deviation, minimum, and maximum of the variable. Clicking **Options** will allow you to disable any of the aforementioned statistics, or enable sum, variance, range, standard error of the mean (S.E. mean), kurtosis, and skewness. You can also choose how you want the output to be organized:

- **Variable list** will print the variables in the same order that they are specified in the Descriptive window.
- **Alphabetically** will arrange the variables in alphabetical order.
- **Ascending means** will order the output so that the variables with the smallest means are first and the variables with the largest means last.
- **Descending means** will order the output so that the variables with the largest means are first and the variables with the smallest means are last



DESCRIPTIVE STATISTICS ON MICROSOFT EXCEL

Variable	Mean	Standard Deviation	Minimum	Maximum	Sum of Squares	Range	Standard Error of the Mean	Kurtosis	Skewness
1	1.00	1.00	1.00	1.00	1.00	0.00	1.00	3.00	0.00
2	2.00	1.00	1.00	3.00	4.00	2.00	1.00	3.00	0.00
3	3.00	1.00	1.00	4.00	9.00	3.00	1.00	3.00	0.00
4	4.00	1.00	1.00	5.00	16.00	4.00	1.00	3.00	0.00
5	5.00	1.00	1.00	6.00	25.00	5.00	1.00	3.00	0.00
6	6.00	1.00	1.00	7.00	36.00	6.00	1.00	3.00	0.00
7	7.00	1.00	1.00	8.00	49.00	7.00	1.00	3.00	0.00
8	8.00	1.00	1.00	9.00	64.00	8.00	1.00	3.00	0.00
9	9.00	1.00	1.00	10.00	81.00	9.00	1.00	3.00	0.00
10	10.00	1.00	1.00	11.00	100.00	10.00	1.00	3.00	0.00

Measures of Position

A measure of position determines the position of a single value in relation to other values in a sample or a population data set. Unlike the mean and the standard deviation, descriptive measures based on quantiles are not sensitive to the influence of a few extreme observations. For this reason, descriptive measures based on quantiles are often preferred over those based on the mean and standard deviation. Quantiles are cut points dividing the range of the data into contiguous intervals with equal probabilities. Certain quantiles are particularly important: The median of a data set divides the data into two equal parts: the bottom 50% and the top 50%. Quartiles divide the data four equal parts and percentiles divide it into hundredths, or 100 equal parts. Note that the median is also the 50th percentile. Deciles divide a data set into tenths (10 equal parts), and the quintiles divide a data set into fifths (5 equal parts). There is always one less quantile than the number of groups created (e.g. There are 3 quartiles dividing the data into 4 equal parts!)

$$\text{QUARTILES } (Q_n): \text{Group data: } \text{LM} + \left(\frac{\sum f b}{\sum f m} \right) c$$

Where: LM = lower class boundaries of the quantitative class

N = total sum of frequencies

$\sum f\pi$ = sum of frequencies before the median frequency

 f_m = frequency of the quartile class

C = class size

n = the n th position

Example: Calculate the (a) 3rd and (b) 1st quartiles of the weight of tomatoes in a can.

Class	Frequency	Cumulative frequency	Class Boundaries
15-18	2	2	14.5-18.5
19-22	5	7	18.5-23.5
24-28	11	18	23.5-28.5
29-33	6	24	28.5-33.5
34-38	3	27	33.5-38.5
39-43	3	30	38.5-43.5
	30		

Solution:

(a) nth position: $\left(\frac{nN}{4}\right)th = \frac{3 \times 30}{4} = \frac{90}{4} = 22.5^{th}$,

Data: $Lm = 28.5$, $N = 30$, $\Sigma ffb = 18$, $fm = 6$, $C = 5$, $n = 3$

$$Q_n = \text{Im} + \left(\frac{\frac{nN}{4} \Sigma f^2 b}{f_m} \right) C,$$

$$Q_3 = 28.5 + \left(\frac{\frac{90}{4} - 18}{6} \right) 5$$

$$= 28.5 + 3.33 = 31.33$$

(b) 1st Quartile

$$\left(\frac{nN}{4}\right)th = \frac{1 \times 30}{4} = \frac{30}{4} = 7.5^{th}$$

Data: $lm = 23.5, N = 30, \Sigma fb = 7, fm = 5, C = 5, n = 1$

$$Q_n = lm + \left(\frac{\frac{nN}{4} - \Sigma fb}{fm}\right) c,$$

$$Q_1 = 23.5 + \left(\frac{\frac{30}{4} - 7}{5}\right) 5$$

$$= 23.5 + 0.5 = 24$$

Example: Determine the (a) 4th and (b) 7th Decile of the average monthly account balance of 100 customer

Naïra	30-34	35-39	40-44	45-49	50-54	55-59	60-64	65-69	70-74	75-79	80-84	85-89
Freq. (f)	1	2	3	8	12	19	26	13	9	4	1	2
CF	1	3	6	14	26	45	71	84	93	97	98	100

(a) nth Position: $\left(\frac{nN}{10}\right)th = \frac{4 \times 100}{10} = \frac{400}{10} = 40^{th}$

Data: $lm = 54.5, \Sigma fb = 26, N = 100, fm = 19, c = 5$

$$D_4 = 54.5 + \left(\frac{4 \times 100 - 26}{19}\right) 5 = 54.5 + \left(\frac{40 - 26}{19}\right) 5 = 54.5 + \left(\frac{14}{19}\right) 5 = 54.5 + 3.68 = 58.19$$

(b) nth Position: $\left(\frac{nN}{10}\right)th = \frac{7 \times 100}{10} = \frac{700}{10} = 70^{th}$ $D_n = lm + \left(\frac{\frac{nN}{10} - \Sigma fb}{fm}\right) c$

Data: $lm = 59.5, \Sigma fb = 45, N = 100, fm = 26, c = 5$

$$D_7 = 59.5 + \left(\frac{7 \times 100 - 45}{26}\right) 5 = 59.5 + \left(\frac{70 - 45}{26}\right) 5 = 59.5 + \left(\frac{25}{26}\right) 5 = 59.5 + 2.69 = 62.19$$

$$= LM + \left(\frac{\frac{nN}{10} - \Sigma fb}{fm}\right) c = 59.5 + \left(\frac{55.5 - 45}{26}\right) 5 = 59.5 + 2.02 = 61.52$$

Example: Use the data in the table below and determine (a) 35th and (b) 90th Percentile

Age	Freq.	CF	Class boundaries
1-5	20	20	0.5-5.5
6-10	30	50	5.5-10.5
11-15	50	100	10.5-15.5
16-20	120	220	15.5-20.5
21-25	80	300	20.5-25.5
26-30	80	380	25.5-30.5
31-35	80	380	30.5-35.5

(a) nth Position: $\left(\frac{nN}{100}\right)th = \frac{35 \times 380}{100} = \frac{13300}{100} = 133^{th}$ $D_n = lm + \left(\frac{\frac{nN}{100} - \Sigma fb}{fm}\right) c$

Data: $lm = 15.5, \Sigma fb = 100, N = 380, fm = 120, c = 5, n = 35$

$$P_{35} = 15.5 + \left(\frac{35 \times 380 - 100}{120}\right) 5 = 15.5 + \left(\frac{133 - 45}{120}\right) 5 = 15.5 + \left(\frac{88}{120}\right) 5 = 15.5 + 1.73 = 22.83$$

(b) nth Position: $\left(\frac{nN}{100}\right)th = \frac{70 \times 380}{100} = \frac{26600}{100} = 266^{th}$

Data: $lm = 25.5, \Sigma fb = 220, N = 380, fm = 80, c = 5, n = 70$

$$P_{35} = 25.5 + \left(\frac{70 \times 380}{100} - 220 \right) \frac{10}{80} = 25.5 + \left(\frac{266 - 220}{80} \right) 10 = 25.5 + \left(\frac{46}{80} \right) 10 = 25.5 + 5.75 = 31.25$$

ANALYSIS OF VARIANCE

Analysis of variance (ANOVA) is an analysis tool used in statistics that splits an observed aggregate variability found inside a data set into two parts: systematic factors and random factors. The systematic factors have a statistical influence on the given data set, while the random factors do not. Analysis use the ANOVA test to determine the influence that independent variables have on the dependent variable in a regression study.

CHI-SQUARE TEST (χ^2)

A chi-square (χ^2) statistic is a measure of the difference between the observed and expected frequencies of the outcomes of a set of events or variables. Chi-square is useful for analyzing such differences in categorical variables, especially those nominal in nature.

When analysis of categorical data is concerned with more than one variable, two-way table (also known as contingency tables) are employed. These tables provide a foundation for statistical inference, where statistical tests question the relationship between the variables on the basis of the data observed.

$$\text{Mathematically } \chi^2 = \sum \frac{(O-E)^2}{E}$$

Observed Values can be determined as

$$O_{ij} = \frac{(O_i \times O_j)}{0}$$

Example: Use the data in the table below and determine the chi-square value

Observed Values (O)

	2018	2019	2020	$O_{.j}$
Windows	20	15	10	45
XP	35	40	25	100
W7	10	20	30	60
W8	15	20	25	60
$O_{i.}$	80	95	90	265 = $O_{..}$

Expected Values (E)

	2018	2019	2020	
Windows	13.58	16.13	15.28	
XP	30.19	35.85	33.96	
W7	18.11	21.51	20.38	
W8	18.11	21.51	20.38	

$$E_{20} = \frac{45 \times 80}{265} = 13.58, E_{15} = \frac{45 \times 95}{265} = 16.13, E_{10} = \frac{45 \times 90}{265} = 15.28, E_{35} = \frac{100 \times 80}{265} = 30.19, E_{40} = \frac{100 \times 95}{265} = 35.85, E_{25} = \frac{100 \times 90}{265} = 33.96$$

$$E_{10} = \frac{60 \times 80}{265} = 18.11, E_{20} = \frac{60 \times 95}{265} = 21.51, E_{30} = \frac{60 \times 90}{265} = 20.38, E_{15} = \frac{60 \times 80}{265} = 18.11, E_{25} = \frac{60 \times 95}{265} = 21.51, E_{35} = \frac{60 \times 90}{265} = 20.38$$

$$\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(20-13.58)^2}{13.58} + \frac{(15-16.13)^2}{16.13} + \frac{(10-15.28)^2}{15.28} + \frac{(35-30.19)^2}{30.19} + \frac{(40-35.85)^2}{35.85} + \frac{(25-33.96)^2}{33.96} + \frac{(10-18.11)^2}{18.11} + \frac{(20-21.51)^2}{21.51} + \frac{(30-20.38)^2}{20.38} + \frac{(15-18.11)^2}{18.11} + \frac{(20-21.51)^2}{21.51} + \frac{(25-20.38)^2}{20.38}$$

$$= 3.035 + 0.079 + 1.825 + 0.742 + 0.549 + 2.364 + 3.632 + 0.106 + 4.541 + 3.632 + 0.106 + 4.541 = 25.1651$$

Chi-Square Tabulated

Degree of freedom = (row total - 1)(column total - 1) = (r - 1)(c - 1)

Critical Values = $\alpha = 0.05, 0.1, 0.025$ etc

Therefore Chi-square tabulated = $\chi^2_{\alpha, (r-1)(c-1)}$

Using the example above the chi-square tabulated at critical value of $\alpha = 0.05$

$$\chi^2_{0.05, (4-1)(3-1)} = \chi^2_{0.05, (2)} = \chi^2_{0.05, 2} = 12.592$$

Decision Rule

If Chi-square Calculated, $\chi^2_{calculated} > \chi^2_{tabulated}$, we reject the null hypothesis, otherwise we accept
To interpret the above calculation then 25.1651 > 12.592, we reject the hypothesis

Simple Regression Analysis

Regression analysis is a process that is estimating or predicting the values of an dependent random variable y on the basis of a known measurement of an independent controlled variable x.

Regression is the study of relationship among variables. The procedure of regression analysis involves formulating a mathematical equation to estimate the value of one variable (dependent) given the value of once or more quantitative variable (independent) x. only one independent variable is involved in simple linear regression is concerned with bi-variate distribution, that is distribution of a set of bi-variate data (x, y) can be presented graphically on a scatter diagram.

Solving the two equation simultaneously gives the value of 'a' and 'b' the equation

$$\bar{y} = a + b\bar{x}, \quad \bar{y} = \frac{\sum y}{n}, \quad \bar{x} = \frac{\sum x}{n}, \quad a = \bar{y} - b\bar{x}; \quad b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

Example: Use the data in the table below and fit the regression equation

Salary (Y)	Grade Level (X)	x	y
250	4	16	1000
300	5	25	15000
420	6	36	2520
80	2	4	160
120	3	12.25	420
200	4	10.89	660
220	4	16.81	902
1590	27.9	120.95	20662
	$\sum x$		$\sum xy$

$$\bar{y} = \frac{\sum y}{n} = \frac{1590}{7} = 227.14, \quad \bar{x} = \frac{\sum x}{n} = \frac{27.9}{7} = 3.99$$

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} = \frac{(20662) - (27.9)(1590)}{7(120.95) - (27.9)^2} = \frac{144634 - 44361}{846.65 - 778.41} = \frac{100273}{68.24} = 1469.42$$

$$a = \bar{y} - b\bar{x} = 227.14 - 1469.42 * 3.99 = 227.14 - 5862.99 = -5635.8$$

$$\text{Regression equation } y = a + bx: y = -5635.85 + 1469.42x$$

SIMPLE CORRELATION ANALYSIS

A simple correlation analysis is an analysis that is concerned with a measure of the relationship between two or more variable x and y, and denoted by p. the variable x and y are assumed to have a joint probability distribution f(x, y).

It is important to remember that the correlation coefficient between two variables is a measure of their linear relationship and a value of p = 0 implies a lack of linearity and not a lack of association. Hence if a strong positive or negative relation exist among them, we will still obtain a zero correction to indicate a non-linear relationship

The measure p of linear relationship between two variables x and y is estimated by the sample correlation coefficient r, where

$$r = \frac{n \sum_{i=1}^n x_i y_i - (\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n \sum x^2 y - \sum x \sum y} = \frac{\sqrt{[n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2][n \sum_{i=1}^n y_i^2 - (\sum_{i=1}^n y_i)^2]}}{\sqrt{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}}$$

Example: Using the data below determine the correlation coefficient and hence draw the scatter diagram.

x	y	x ²	y ²	xy
1	6	1	36	6
2	4	4	16	8
3	3	9	9	9
4	5	16	25	20
5	4	25	16	20
6	2	36	4	12
21	24	91	106	258

$$\sum x = 21, \sum y = 24, \sum x^2 = 91, \sum y^2 = 106, \sum xy = 258, n = 6$$

$$r = \frac{\sqrt{(6(91) - (21)^2)(6(106) - (24)^2)}}{\sqrt{(546 - 441)(636 - 576)}} = \frac{\sqrt{(105)(60)}}{63} = \frac{-54}{63} = -0.008$$

The result indicates very weak negative (-0.008) correlation between x and y variables.

Example: Determine the correlation value and hence interpret your result

x(Mathematics)	y(Physics)	xy	x ²	y ²	Data
4	3	12	16	9	$\sum xy = 203$
8	9	72	64	81	$\sum y = 36$
6	8	48	36	64	$n = 7$
5	7	35	25	49	$\sum x = 33$
3	2	6	9	4	$\sum x^2 = 179$
2	1	2	4	1	$\sum y^2 = 244$
5	6	30	25	36	
33	36	203	179	224	

$$R = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}} = \frac{7(203) - (33)(36)}{\sqrt{(164)(412)}} = \frac{233}{\sqrt{67568}} = \frac{233}{259.94} = 0.896$$

Result: There is a strong positive correlation between the performance of student in physics and mathematics this shows that a student is performing well in mathematics the student can as well perform well in physics

Exercise

- a) The grades of a class of a student on a mid-term report (x) and on the final examination (y) are as follows

x	y	x ²	y ²	xy
77	50	71	72	81
82	66	78	34	47
				85
				96
				99
				67
				68

Determine the correlation coefficient.

- b) Calculate the co-efficient of the correlation to show the association between these two steps if marks obtained out of 10.

Reading (x)	4	6	5	5	4	6	6
arithmetic(y)	3	-3	3	3	2	3	2

Spearman's Rank Correlation

Spearman's rank correlation measures the strength and direction of association between two ranked variables. It basically gives the measure of monotonicity of the relation between two variables i.e. how well the relationship between two variables could be represented using a monotonic function.

The formula for Spearman's rank coefficient is:

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

ρ = Spearman's rank correlation coefficient

d = Difference between the two ranks of each observation

n = Number of observations

The Spearman Rank Correlation can take a value from +1 to -1 where,

- ✓ A value of +1 means a perfect Positive association of rank
- ✓ A value of 0 means that there is no association between ranks
- ✓ A value of -1 means a perfect negative association of rank

The grades of a class of a student on a mid-term report (x) and on the final examination (y) are as follows

x	77	50	71	72	81	84	86	89	67
y	82	66	78	34	4	85	95	99	68
R_x	5	1	3	4	5	7	8	9	2
R_y	6	3	5	1	2	7	8.5	8.5	4
$d = R_x - R_y$	-1	-2	-2	3	3	0	-0.5	0.5	-2
d^2	2	4	4	9	9	0	0.25	0.25	4
$4-39.5 = \sum d^2$									

$$\rho = 1 - \frac{6(39.5)}{9(9^2 - 1)} = 1 - \frac{237}{9(80)} = 1 - \frac{237}{720} = 1 - 0.329 = 0.671$$

Conclusion: The Result ($\rho = 0.67$) indicates that there is a strong positive association between the mid-term report and the final examination.

WILCOXON RANK SUM TEST

The Wilcoxon test, which can refer to either the rank sum test or the signed rank test version, is a nonparametric statistical test that compares two paired groups. The tests essentially calculate the difference between sets of pairs and analyze these differences to establish if they are statistically significantly different from one another.

The Wilcoxon rank sum test can be used to test the null hypothesis that two populations have the same continuous distribution. A null hypothesis is a statistical test that says there's no significant difference between two populations or variables.

Example: The table below shows the hours of relief provided by two analgesic drugs in 12 patients suffering from arthritis. Is there any evidence that one drug provides longer relief than the other?

Cases	Drugs A	Drugs B	DrugB-DrugA	Diff	Rank	Sign
1	2.0	3.5	+1.5	0.1	1	-
2	3.6	5.7	+2.1	0.2	2	-
3	2.6	2.9	+0.3	0.3	3	+
4	2.6	2.4	-0.2	0.6	4	-
5	7.3	9.9	+2.6	1.5	5.5	+
6	3.4	3.3	-0.1	1.5	5.5	+
7	14.9	16.7	+1.8	1.8	7	+
8	6.6	6.0	-0.6	2.0	8	+
9	2.3	3.8	+1.5	2.1	9	+
10	2.0	4.0	+2.0	2.3	10	+
11	6.8	9.1	+2.3	2.6	11	+
12	8.5	20.9	+12.4	12.4	12	+

$$W^- = 1 + 2 + 4 = 7, W^+ = 3 + 5.5 + 5.5 + 7 + 8 + 9 + 10 + 11 + 12 = 71$$

$$n = \frac{n(n+1)}{2} = \frac{12(12+1)}{2} = \frac{12(13)}{2} = \frac{156}{2} = 78$$

$$W = \max(W^-, W^+) = 71$$

The critical value of W can be found in the table of critical values. To determine the appropriate critical value from Table we need sample size ($n=12$) and our two-sided level of significance ($\alpha=0.05$). The critical value for this two-sided test with $n=12$ and $\alpha=0.05$ is 14 and the decision rule is as follows: Reject H_0 if $W \leq 14$.

In this example, $W^- = 1 + 2 + 4 = 7$, $W^+ = 3 + 5.5 + 5.5 + 7 + 8 + 9 + 10 + 11 + 12 = 71$, $W = 71$ and $W^- = 7$. Recall that the sum of the ranks (ignoring the signs) will always equal $n(n+1)/2$. As a check on our ranks, we have $n = \frac{n(n+1)}{2} = \frac{12(12+1)}{2} = \frac{156}{2} = 78$ which is equal to $71+7$. The test statistic is $W = 7$.

Conclusion: We reject H_0 because $7 < 14$. Therefore, we have statistically significant evidence at $\alpha=0.05$, that the drugs B provides more relief for arthritis Patients than Drugs A.

Assignment: A study is run to evaluate the effectiveness of an exercise program in reducing systolic blood pressure in patients with pre-hypertension (defined as a systolic blood pressure between 120-139 mmHg or a diastolic blood pressure between 80-89 mmHg). A total of 15 patients with pre-hypertension enroll in the study, and their systolic blood pressures are measured. Each patient then participates in an exercise training program where they learn proper techniques and execution of a series of exercises. Patients are instructed to do the exercise program 3 times per week for 6 weeks. After 6 weeks, systolic blood pressures are again measured.

The data are shown below.

Patient Systolic Blood Pressure		
Patients	Before Exercise	After Exercise
1	125	118
2	132	134
3	138	130
4	120	124
5	125	105
6	127	130
7	136	130
8	139	132
9	131	123
10	132	128
11	135	126
12	136	140
13	128	135
14	127	126
15	130	132

Is there a difference in systolic blood pressures after participating in the exercise program as compared to before?

KRUSKAL WALLIS TEST (H)

A popular nonparametric test to compare outcomes among more than two independent groups is the Kruskal Wallis test. The Kruskal Wallis test is used to compare medians among k comparison groups ($k > 2$) and is sometimes described as an ANOVA with the data replaced by their ranks. The null and research hypotheses for the Kruskal Wallis nonparametric test are stated as follows:

H_0 : The k population medians are equal versus

H_1 : The k population medians are not all equal Mathematically

$$H = \frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} - 3(N+1)$$

$$W^- = 1 + 2 + 4 = 7, W^+ = 3 + 5.5 + 5.5 + 7 + 8 + 9 + 10 + 11 + 12 = 71$$

$$n = \frac{n(n+1)}{2} = \frac{12(12+1)}{2} = \frac{12(13)}{2} = \frac{156}{2} = 78$$

$$W = \max(W^-, W^+) = 71$$

The critical value of W can be found in the table of critical values. To determine the appropriate critical value from Table we need sample size ($n=12$), and our two-sided level of significance ($\alpha=0.05$). The critical value for this two-sided test with $n=12$ and $\alpha=0.05$ is 14 and the decision rule is as follows: Reject H_0 if $W \leq 14$. In this example, $W^- = 1 + 2 + 4 = 7, W^+ = 3 + 5.5 + 5.5 + 7 + 8 + 9 + 10 + 11 + 12 = 71$. $W^+ = 71$ and $W^- = 7$. Recall that the sum of the ranks (ignoring the signs) will always equal $n(n+1)/2$. As a check on our ranks, we have $n = \frac{n(n+1)}{2} = \frac{12(12+1)}{2} = \frac{156}{2} = 78$ which is equal to $71+7$. The test statistic is $W = 7$.

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H_0 : The k population medians are equal versus

H_a : The k population medians are not all equal

Mathematically

$$H = \frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} - 3(N+1)$$

The procedure for the test involves pooling the observations from the k samples into one combined sample, keeping track of which sample each observation comes from, and then ranking lowest to highest from 1 to N , where $N = n_1 + n_2 + \dots + n_k$.

Example: A clinical study was carried at in an IDP Camps in Maiduguri to examine the level of health development among children. Three diets are compared, ranging from 5%, 10% and 15% protein. The improvement level of the children that participants following each diet are shown below:

	5% Protein	10% Protein	15% Protein
3.1	3.8	4.0	
2.6	4.1	5.5	
2.9	2.9	5.0	
	3.4	4.8	
	4.2		

Is there any difference in health improvement levels among children on the three different Protein levels?

Solution

Hypotheses

H_0 : The three population medians are equal versus

H_1 : The three population medians are not all equal

Rank 5%	Rank 10%	Rank 15%
4	6	7
1	8	12
2.5	2.5	11
	5	10
	9	
7.5	30.5	40

Test Statistic for the Kruskal Wallis Test

The test statistic for the Kruskal Wallis test is denoted H and is defined as follows:

$$H = \left(\frac{12}{M(M+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right) - 3(N+1)$$

$$H = \left(\frac{12}{M(M+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right) - 3(N+1) = \frac{12}{12(13)} \left(\frac{7.5^2}{3} + \frac{30.5^2}{5} + \frac{40^2}{4} \right) - 3(13) = 7.52$$

Conclusion: To determine the appropriate critical value we need sample sizes ($n_1=3$, $n_2=5$ and $n_3=4$) and our level of significance ($\alpha=0.05$). For this example the critical value is 5.656, thus we reject H_0 because $7.52 \geq 5.656$, and we conclude that there is a difference in median health improvement among the three different protein level administered on the children in the IDP camps.

Assignment: A computer analyst is interested in studying the speed of the computers operating systems which has the records in the table below:

Windows XP	Windows 7	Windows 8	Windows 10
1.8	4.1	4.7	5.5
1.5	3.8	5.3	6.9
2.5	2.9	6.2	7.2

3.3	1.3	4.8	
1.0	4.2		
	5.2		

Is there any difference in the median of the processor speed of the Operating Systems? $\alpha = 0.05$

ANALYSIS OF VARIANCE (ANOVA)

Analysis of Variance (ANOVA) is a statistical tool used for the comparison of several treatment means. It therefore provides:

1. A subdivision of the total variation between the experimental units into separate components, with each component depicting a different source of variation
2. An estimate of the underlying variation which provides a basis for inferences about the effect of the applied treatment.

ONE WAY ANOVA TABLE

Source of variation	Sum of Squares	Degree of Freedom	Mean Squares	Ratio	$\frac{MS_{\text{treatment}}}{MS_{\text{error}}}$
Between Treatment	$\sum_{i=1}^t \left(\frac{T_i^2}{n_i} \right) - \frac{T^2}{n}$	$t - 1$	$\frac{SS_{\text{treatment}}}{t - 1}$	$\frac{t - 1}{t - 1}$	
Error	$S - \sum_{i=1}^t \left(\frac{T_i^2}{n_i} \right)$	$n - 1$			
Total	$S - \frac{T^2}{n}$	$n - 1$			

Hypothesis test to determine whether there is a treatment difference is given as

$$H_0: T_i = \delta \text{ versus } H_0 \neq \delta$$

Decision Rule:

We compare $F = \frac{MS_{\text{treatment}}}{MS_{\text{error}}}$ with $F(t - 1, n - 1)$ at a level of significance to decide whether or not to reject

H_0

We reject H_0 if $F_{\text{calculated}} > F_{\text{tabulated}}$

Example

Computers HP, Dell and Samsung were infected with trojan virus. The essence of the experiment was to compare the effect of the virus on the intel processor speed, as shown below:

HP	17.1	4.9	23.6	16.0	18.7	21.7	21.4	23.0	19.0	18.6
Dell	24.3	21.1	22.1	26.4	23.2	20.7	22.9	19.6	22.4	
Samsung	14.3	14.5	17	13.2	15.6	10.3				

Use the data above to test whether there is a difference in the effects of the virus.

SOLUTION

$$T_1 = 194, T_2 = 203.5, T_3 = 85, n_1 = 9, n_2 = 6$$

$$\text{The Correction Factor } \frac{T^2}{n} = \frac{(T_1 + T_2 + T_3)^2}{n_1 + n_2 + n_3} = \frac{(194 + 203.5 + 85)^2}{25}$$

$$S = \sum y_i^2 = 17.1^2 + 14.9^2 + \dots + 10.3^2 = 9705.41$$

$$SS_{\text{total}} = 9705.41 - 9312.25 = 393.16$$

$SS_{\text{treatment}} =$

$$\sum \left(\frac{T_i^2}{n_i} \right) - \frac{T^2}{n} = \left(\frac{194^2}{10} + \frac{203.5^2}{9} + \frac{85^2}{6} \right) - 9312.25 = 9569.13 - 9312.25 = 256.88$$

Theophilus

$$SS_{\text{Error}} = 393.16 - 256.88 = 136.28$$

ANOVA Table

Source	SS	DF	MS	F
Treatment	256.88	2	128.44	20.75
Error	136.28	22	6.19	
Total	393.16	24		

Hypothesis Test:

$$H_0: \tau_i = \sigma \text{ Vs } H_1: H_0 \text{ not true}$$

Test Statistic:

$$F_{\text{calculated}} = 20.75$$

$$F_{\text{tabled}} (2, 22) \text{ at } \alpha = 0.05 \text{ level of significance is } 3.44$$

Since $F_{\text{calculated}} > F_{\text{tabled}}$ (2, 22), we reject H_0 . This implies that there is significant difference between the output of the virus of the affected computers.

Median Test

The median test is a procedure for testing whether two independent groups differ in central tendencies. It gives information as to whether it is likely that two independent groups have been drawn from population with the same median. The null hypothesis states that the two groups considered are from population with the same median.

The following procedure may be followed when carrying out median test:

- Obtain the common median of the samples considered by arranging the data in an order of magnitude.
- Determine the observations that are above, below or equal to the common median.
- Construct 2×2 compared with $\chi^2 = \frac{n(n_1 - n_2)^2}{n_1 n_2}$

This is compared with $\chi^2_{1-\alpha, 1}$ level of significance. We reject H_0 if

$$\chi^2_{\text{calculated}} > \chi^2_{1-\alpha, 1}$$

The table below shows the scores of two groups of student taught by two lecturers in a statistics course. Perform median test on data using common median = 60. Use $\alpha = 0.05$ level of significance for the test.

Scores by Lecturer X	60	60	60	60	50	60	75	65	70	40	35	42	36	67
Scores by Lecturer Y	60	55	64	78	60	48	56	52	77	80				81

Solution:

The 2×2 contingency table is shown below:

	Lecturer X	Lecturer Y	Total
Score above the common median	4	5	9
Score equal to or below the common median	8	6	14
Total	12	11	23

$$\chi^2 = \frac{23(24 - 40)^2}{(123)(115)} = \frac{16632}{5068} = 0.32$$

$$\chi^2_{1-\alpha, 1} = 3.84$$

$$\chi^2 = 0.32 < \chi^2_{1-\alpha, 1} = 3.84, \text{ We do not reject } H_0.$$

15.4 Mann-Whitney U-Test

The test is used to test whether two independent groups have been drawn from the same population. The data in the two samples under consideration are combined and ranked in order of magnitude. The sum of the ranks for each sample is obtained and used to compute the following:

- Mann-Whitney U-statistic

$$U = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

Where n_2 and n_1 are the sample sizes for the two samples and R_1 is the sum of rank for n_1

- The mean of U-statistics is given as

$$\mu_u = \frac{n_1 n_2}{2}$$

iii. The standard deviation of U is given

$$\sigma_u = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$$

iv. The z-score for the U -statistic is given as

$$Z = \frac{U - \mu_u}{\sigma_u}$$

This is compared with the critical value from normal distribution table at α level of significant. The data below show the time taken (in seconds) by participants to answer a question at the MTN competition: "Who Wants to be a Millionaire" in the first two weeks of March 2010.

Participants (1 st Week)	9	12	15	18	20							
Participants (2 nd Week)	22	24	18	16	30	35						

Test for the null hypothesis that the median of the time taken by participants in the first week is equal to or less than the median of the median of the time by participant in the second week.

Solution:

Ranks of time taken by Participants (1 st week)												
Ranks of time taken by Participants (2 nd week)												

Sum of Rank (R_1) = 17, n_1 = 5,

Sum of Rank (R_2) = 61, n_2 = 7

$$U = (5)(7) + \frac{5(5+1)}{2} - 17 = 35 - 2 = 33$$

$$\mu_u = \frac{(5)(7)}{2} = 17.5$$

$$Z = \frac{U - \mu_u}{\sigma_u} = \frac{33 - 17.5}{\sqrt{\frac{5(5+1)(7+1)}{12}}} = \frac{15.5}{6.16} = 2.51$$

The critical value from standard normal distribution is 1.64.

Decision: We reject H_0 . Since $2.51 > 1.64$.

STUDY